

Ashutosh Modi

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SUMMARY

Backend/platform engineer with **3.5+ years** building telecom-scale distributed systems in **C/C++**, **Python**, **SQL**, and **Linux**. Designed **gRPC/Protobuf** migration frameworks, service-discovery/orchestration runtimes, compiler automation, and internal AI developer tooling that improve reliability, modernization speed, and developer productivity.

EXPERIENCE

Amdocs

Sep 2022 – Present

Software Developer — Telecom Backend Platform

Gurgaon, India

- Designed and shipped a **Tuxedo → gRPC migration framework** for Tier-1 telecom backends, preserving **every line of business logic**; validated as a production-service POC and now rolling out across CK Hutchison's **Three UK** and **Three Ireland** to remove a multi-million-dollar Oracle dependency.
- Drove day-to-day technical execution for a **4–5 engineer team** on a Tier-1 telecom modernization program since **Dec 2025** – code review, unblocking, and the engineering manager's primary technical interface (informal capacity).
- Replaced **TIBCO WebFOCUS** (formerly Information Builders) reporting with an in-house Python library and Lex/Yacc auto-converter for **500+** legacy scripts, eliminating **millions in annual licensing per client deployment**.
- Hardened a long-lived C/C++ codebase: guard-clause refactors cut potential vulnerabilities by **30%**, a build-time prototype generator catches **80%** of would-be production defects, and resolved **95%** of legacy compile-time warnings.
- Owned company-wide **Cursor** rollout as the AI development platform plus reusable Vim / VS Code snippets and Shell automation, cutting team toolchain friction by **50%+**; operate Oracle DB on large telecom datasets.

SELECTED PROJECTS (AMDOCS)

Tuxedo → gRPC Migration Framework | *gRPC, Protobuf, HAProxy, Consul, Nomad*

Jun 2025 – Nov 2025

- Built a generic protocol shim translating Tuxedo's flattened buffers into strongly-typed Protobuf with **zero changes to business code**.
- HAProxy + Consul + Nomad** runtime handles load balancing, service discovery, workload scheduling, and config sync for incremental rollout.

Internal AI Development Platform | *Cursor, Agent Workflows, Knowledge Engineering*

Aug 2025 – Present

- Authored the team's knowledge base, agent skills, and workflow rules from scratch; built end-to-end agent workflows for git ops, PR creation, and recurring code review.
- Engineered a **layered context-loading strategy** that lets agents reason over multi-MB telecom codebases while preserving project conventions.

XFocus | *Python, Pandas, NumPy, Lex/Yacc*

Jul 2024 – Oct 2024

- Built a declarative Python library on Pandas/NumPy that replaces **TIBCO WebFOCUS** for telecom business reporting, eliminating millions in annual licensing per client deployment.
- Authored a Lex/Yacc compiler that auto-converts legacy WebFOCUS scripts to the new library, cutting per-script migration time from **48 hours → 40 minutes** across **500+** scripts.

C/C++ Prototype Generator | *Lex/Yacc, Shell*

Jun 2023 – Sep 2023

- Built a Lex/Yacc compiler that scans every C source at build time and emits function prototypes across **500+** files.
- Wired the generated prototypes through preprocessor **#if** guards, catching bad merges, incorrect calls, and type mismatches at compile time.

TECHNICAL SKILLS

Languages: C/C++, Python, Rust, SQL, Linux Shell Scripting

Distributed Systems: gRPC, Protobuf, HAProxy, HashiCorp Consul, HashiCorp Nomad, Consul Template, Kafka

Tooling & Databases: Lex/Yacc, Git/GitHub, Vim, VS Code, Tmux, Linux/UNIX, Oracle DB, Cursor

Concepts: Distributed Systems, Microservices, Service Discovery, Load Balancing, High Availability, Knowledge Engineering, AI Agent Workflows, Context Design, Data Structures & Algorithms, System Design

EDUCATION

Guru Nanak Dev University

2018 – 2022

Bachelor of Technology in Computer Science — CGPA: 8.0 / 10.0

Amritsar, India

AWARDS & RECOGNITION

Pinnacle Performer, Amdocs – Feb 2026 | **Employee of the Month**, Amdocs – Feb 2025 & Feb 2026

Placement Head, Guru Nanak Dev University | **Winner**, Google Workshop competition